

Concrete Compressive Strength Prediction: A Comparative Study of Eleven Machine Learning Models with Explainability Analysis

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Abstract

This report presents a comprehensive machine learning pipeline for predicting the compressive strength of concrete using the UCI Concrete Compressive Strength dataset [1]. We evaluate eleven baseline algorithms, perform hyperparameter tuning with Optuna on the five most promising models, and interpret the final model's predictions using SHAP. The best-performing model, CatBoost, achieves a test R^2 of **0.9324** with an RMSE of **4.0119 MPa** and a generalisation gap of only **0.00164**, demonstrating robust predictive performance.

1 Introduction

Accurate prediction of concrete compressive strength is critical for structural engineering and quality control. Traditional empirical models are often limited to specific mix designs. Machine learning (ML) offers a data-driven alternative that can capture non-linear relationships among the ingredients and curing age. This project aims to build a robust ML pipeline that not only delivers accurate predictions but also provides interpretability through SHAP (SHapley Additive exPlanations). The work is part of a skills-demonstration portfolio; no novel algorithms are proposed, but the implementation follows best practices in data splitting, scaling, cross-validation, tuning, and explainability using libraries such as Scikit-learn, XGBoost, LightGBM, and CatBoost.

2 Dataset and Preprocessing

The UCI Concrete Compressive Strength dataset [1] contains 1,030 instances, with eight numerical input features: cement, blast furnace slag, fly ash, water, superplasticizer, coarse aggregate, fine aggregate, and age. The target variable is the concrete compressive strength, measured in MPa. An additional feature, the water-cement ratio, was engineered from the cement and water contents. SHAP analysis identified the water-cement ratio as one of the most influential features in predicting compressive strength.

After removing duplicates and inspecting outliers (retained as legitimate mix variations), the data were split into training (80%) and testing (20%) sets using a fixed random seed (seed=2006). Features were standardised using a 'StandardScaler' fitted only on the training data; the scaled versions were used for distance-based models (SVR, KNN), while tree-based models used the original unscaled values.

3 Methodology

3.1 Baseline Models

Eleven models were trained with default hyperparameters to establish a performance baseline:

- Linear models: Linear Regression, Ridge, Lasso
- Distance-based: K-Nearest Neighbours, Support Vector Regression (SVR)
- Tree-based: Decision Tree, Random Forest, Gradient Boosting, XGBoost, LightGBM, CatBoost.

Each model was evaluated on the test set using R^2 , RMSE, and MAE, and its generalisation capability was assessed via 5-fold cross-validated R^2 on the training data.

3.2 Hyperparameter Tuning with Optuna

The five best baseline models—CatBoost, LightGBM, XGBoost, Random Forest, and Gradient Boosting—were further tuned using Optuna with the TPE sampler. For each model, we conducted 70 trials to maximise the validation R^2 . For boosted models, an additional 20% validation split was held out from the training data to enable early stopping; for bagging models, 5-fold cross-validation was used inside the objective. After identifying the best hyperparameters, each model was retrained on the full training set.

3.3 Explainability with SHAP

The final selected model (CatBoost) was interpreted using the ‘TreeExplainer’ from the SHAP library. Global feature importance was visualised with a beeswarm summary plot, dependence plots for the two most influential features (age and water-cement ratio) were generated.

4 Results and Discussion

4.1 Model Performance

Table 1 reports the test performance of the top five models (including the baseline CatBoost that outperformed its tuned version). All results are from the held-out test set. The ‘Gap’ column is defined as the difference between the cross-validation R^2 (or validation R^2 for boosted models) and the test R^2 ; a smaller gap indicates better generalisation.

Table 1: Performance comparison of the best models (sorted by test R^2).

Model	Type	R^2	RMSE (MPa)	Gap
CatBoost	Baseline	0.9324	4.0119	0.00164
GradientBoosting	Tuned	0.9295	4.0976	0.00536
XGBoost	Tuned	0.9203	4.3578	-0.0169*
LightGBM	Baseline	0.9193	4.3853	0.00534
RandomForest	Tuned	0.8988	4.9114	0.00589

The baseline CatBoost achieved the highest test R^2 with a very small generalisation gap, outperforming even its tuned version (which had $R^2 = 0.9304$ but a much larger gap). GradientBoosting performed very closely with an even smaller gap, but CatBoost was ultimately selected for its simplicity and superior accuracy. Figure 1 visualises the R^2 for all models, confirming the top position of CatBoost baseline.

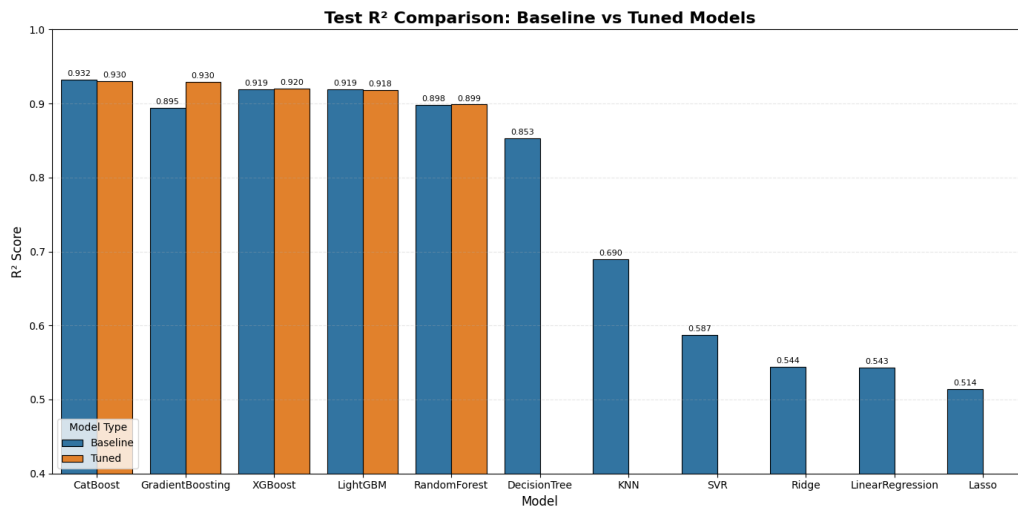


Figure 1: Comparison of test R^2 scores across the evaluated models.

4.2 SHAP Explainability

Figure 2 presents the global feature importance (beeswarm plot) for the CatBoost baseline model. The most influential features are **age**, **water-cement ratio**, and **blast furnace slag**, which aligns with domain knowledge: longer curing age and lower water-cement ratio typically increase strength.

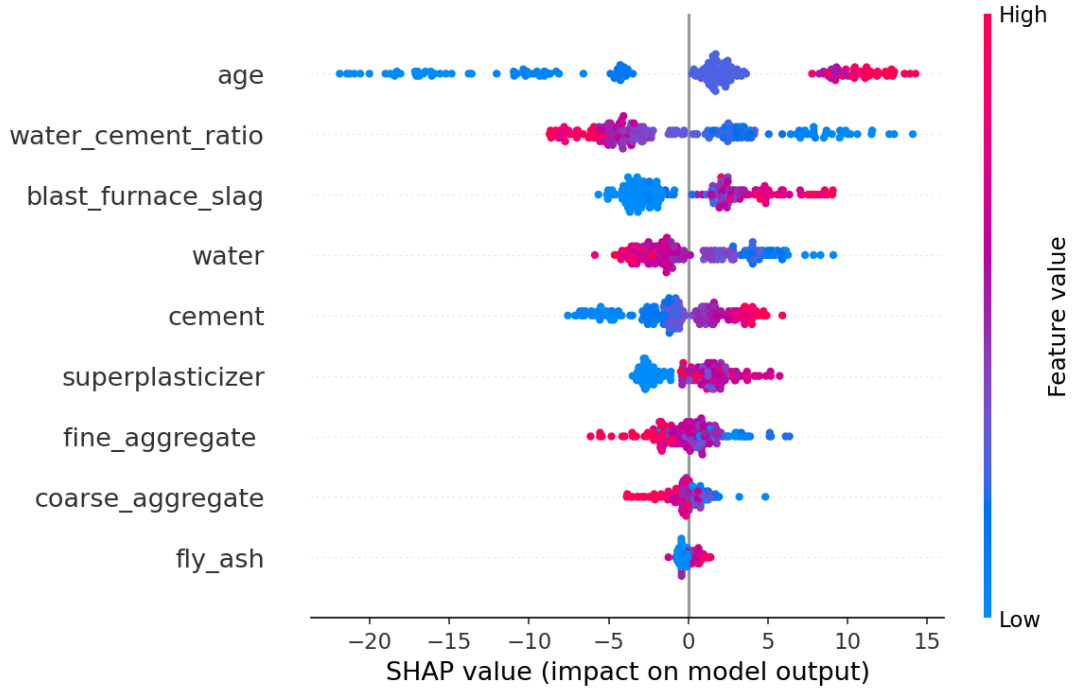


Figure 2: Global SHAP summary (beeswarm) for the final CatBoost model.

To better understand the impact of the two most important features, we generated dependence plots (Figures 3a and 3b).

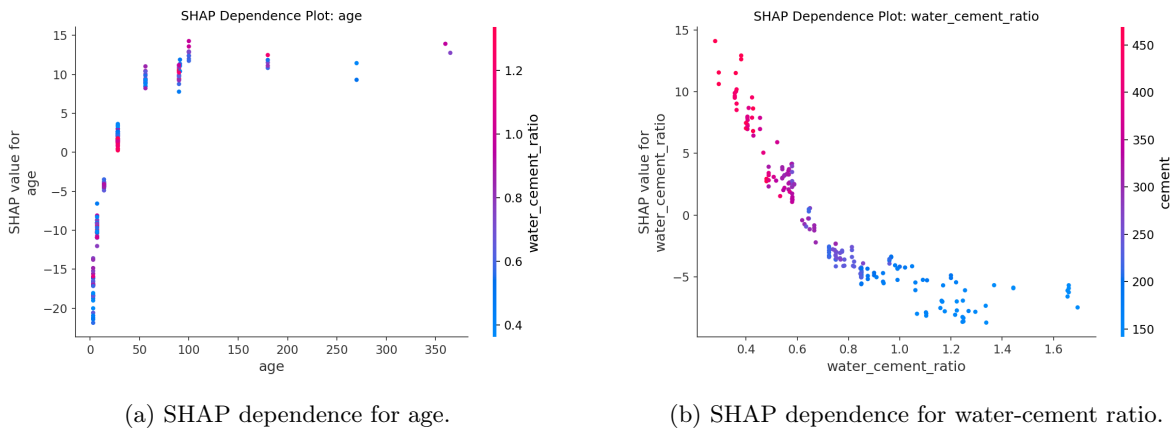


Figure 3: Dependence plots for the two most influential features.

The dependence plots reveal clear monotonic relationships. For age (Figure 3a), the SHAP values increase steadily with age, indicating that longer curing times consistently contribute to higher predicted strength. For the water-cement ratio (Figure 3b), the SHAP values decrease as the ratio increases, confirming that higher water content relative to cement reduces compressive strength. The colour scale in both plots represents the value of the other feature (age or water-cement ratio), showing some interaction effects, though the primary trend remains monotonic.

5 Conclusion

This project successfully built and evaluated a comprehensive set of ML models for concrete compressive strength prediction. The CatBoost model with default hyperparameters outperformed all tuned alternatives, achieving a test R^2 of 0.9324 and an RMSE of 4.0119 MPa, with a minimal generalisation gap of 0.00164. The inclusion of SHAP analysis enhances model trustworthiness by revealing that age, water-cement ratio, and blast furnace slag are the dominant drivers of strength. The pipeline demonstrates best practices in data handling, model selection, hyperparameter tuning, and explainability, making it a solid foundation for future work in materials science.

References

- [1] I-Cheng Yeh. Concrete Compressive Strength. UCI Machine Learning Repository, 1998. DOI: <https://doi.org/10.24432/C5PK67>.